Price Prediction - How Much Data Do You Need?

Ablation Study of Short Term Electricity Price Prediction Utilizing Trees, Neural Networks, and AutoGluon

Master's Thesis

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I. Introduction

In a world of increasing amounts of electric vehicles, data centres, and other large-scale electricity-hungry industries, electricity grids are beginning to feel the strain [1], contributing to an increase in electricity prices [2]. When faced with such increased prices, it is beneficial, from a monetary point of view, to schedule electricity intensive tasks during periods where prices are low. However, to do this, one must know when such low-price periods occur. In Denmark, the price of electricity is, in simple terms, fixed for the next 24-hour period at noon daily[1]. This is often sufficient for household electricity use, as one can typically schedule dishwashers, washing machines, and electric vehicle charging to happen during the lowest point of that period. This can be done either through direct scheduling on individual appliances or smart devices, or through smart home management systems such as Home Assistant [4]. Industrial needs for energy can involve larger and more time-sensitive electricity demand [5], and may therefore be more expensive. As an example, one might imagine a near-future where fleets of electric trucks need their batteries charged before delivery. In a scenario like this, it may be beneficial for such companies, to be able to forecast electricity prices further into the future, and plan their electricity consumption accordingly, to minimize charging costs.

When attempting to predict electricity prices, many different aspects can be considered [6]. Among these, the ones of particular interest to this paper are:

- Weather
- Electricity production details
- Electricity consumption details
- Historical- or T-minus data[2]
- Time of year/month/week/day
- Electricity exchange with foreign countries

One factor that influences electricity prices is the weather, due to its direct impact on both electricity production [7] and consumption [8]. In a country that has invested extensively in wind-energy, such as Denmark, the weather directly impacts electricity production, as more that 80% is produced by renewable sources [9]. Steady winds, and to a lesser extent, clear skies therefore affect energy production quite heavily in Denmark, as necessities for efficient wind and solar energy production. When it comes to consumption, human consumption patterns also change depending on the weather [8]. When it is cold, people turn on the heat to warm themselves, and when it is hot, they turn on air conditioning, or fans, to stay cool. Further, when it is cold, such as in the winter months, it is typically also darker, necessitating the need to use lights more frequently.

These factors suggest that one can use the weather to predict electricity production and consumption (P&C) patterns, which implies that weather data may carry useful information for electricity price forecasting. A substantial body of prior work has examined short-term electricity price forecasting using statistical, machine-learning, and

[1] This is a simplification for ease of reading. For actual energy trading details one may visit [3]. It is also further explained in subsection III-C

[2] Data from an earlier point in time before the current time, e.g. 24 hours before.

deep-learning approaches [10, 11, 12], showing various results in different forecasting scenarios. However, many of the reported results are obtained in specific market settings or with highly specialized forecasting architectures, such as in [12]. Much of this prior work shares common limitations, such as results typically being reported for a single model trained on a fixed feature set. This may obscure the answer to what data actually impact the performance of models in general. Someone wishing to deploy a forecasting system is therefore left without clear guidance on whether investing in comprehensive data collection is worthwhile, or whether a simpler feature set would perform comparably. Further, while related studies exist, the Danish setting combines several factors that are not jointly emphasized in most of the literature reviewed here, such as its heavy investment into wind-energy [9], its division of the energy grid into two distinct bidding zones, and its heavy energy-exchange with its neighbouring countries.

This paper addresses these gaps through a large-scale ablation study, systematically varying the feature composition, forecasting horizon, and prediction target across 1.052 trained models. Rather than reporting the best achievable performance under ideal conditions, the goal is to answer a practically motivated question: how much data do you actually need? This is not merely an academic exercise, but a question of practicality. An energy trading company deploying a real-time forecasting system must weigh the cost of acquiring and maintaining each data source against the marginal accuracy it may provide. Understanding which feature categories contribute meaningfully, and at which horizons they stop being useful, has direct commercial value.

To investigate this dynamic across a broad spectrum of architectures, eight different types of general-purpose machine learning algorithms and frameworks are evaluated:

1) RandomForest
2) XGBoost
3) LightGBM
4) CatBoost
5) LSTM
6) GRU
7) Transformer
8) AutoGluon

Model training will be based on weather data gathered from the Danish Meterological Institute (DMI), as well as electricity P&C data from EnergiNet (EN). To comprehensively evaluate the chosen forecasting approaches, the experiments are structured around three distinct prediction settings. First, the models are trained on various data constellations, ranging from purely weather-related data, to a more comprehensive dataset of weather, grid, and historical price data. This is done to isolate the impact of each specific data category. Second, the forecasting horizon is expanded sequentially to evaluate how the accuracy of each algorithm deteriorates as it predicts further into the future. Finally, the approaches are trained using two distinct prediction targets: predicting the absolute electricity price and predicting the price delta. For the purpose of this paper, the price delta refers to the change in price from the current hour to the forecasted hour. This dual-target approach is motivated by a desire to anticipate model overfitting and to investigate whether the algorithms are more adept at capturing relative price fluctuations rather than absolute prices. The overall purpose of this ablation study is to uncover how different machine learning frameworks react when given identical training conditions under these varying settings.

Producing a meaningful ablation study in this setting required substantial preprocessing effort. The raw data sources used for this paper spans nearly 11 years, and consists of station-level weather measurements, energy production and consumption records, and price data across two format changes. This data was not designed for machine learning consumption. Constructing a consistent, temporally aligned training dataset required geographic filtering of weather stations, cross-year column homogenisation, resolution standardisation, and imputation of structurally missing data. This preprocessing pipeline is a necessary foundation for the validity of the experimental results, and is described in detail in section IV to enable reproducibility.

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Related to this, one company has actually already utilised weather data to predict electricity production and consumption in Denmark. This company is called Midas Energy [13]. They do this with the purpose of supplying their customers, energy trading companies, with this information, in order to enable them to make decisions on a stronger foundation. They provide short term energy production and consumption forecasts, so that their customers may buy and sell electricity with the ability to consider the future to a greater degree than they otherwise would be able to [13]. Notably, Midas Energy provides production and consumption forecasts, not price forecasts, but is interested in expanding into such areas, which is why they have acted as partners during the development process of this paper. Their contributions have been mostly related to data acquisition, performance metric evaluations, and benchmarking, and their contributions will be discussed further when relevant.

II. Related Works

Related works goes here.

A. *Separated by subsections*

As stated in title.

III. Problem Formulation & Prerequisite

A. *Problem formulation*

Problem formulation goes here.'

B. *Forecasting Approach Selection*

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As previously mentioned, the forecasting approaches chosen for predicting the electricity prices in Denmark are:.

1) RandomForest (RF)
2) XGBoost (XGB)
3) LightGBM (LGBM)
4) CatBoost (CB)
5) Long-Short-Term-Memory (LSTM)
6) Gated Recurrent Unit (GRU)
7) Transformer
8) AutoGluon (Gluon)

These approaches can be divided into three general groups: The tree-based models (RF, XGB, LGBM, and CB), the Neural network-based models (LSTM, GRU, and Transformer), and Gluon. The tree based models share several similarities, such as their prediction approach being based on decision trees, as well as their fast training times. The neural network-based models were selected due to their inherent memory capabilities (e.g., hidden states and attention mechanisms), which conceptually allow them to capture complex, long-term temporal dependencies in sequential data. However, due to their architectural complexity and the computational expense of sequential processing, they exhibit significantly longer training times compared to tree-based algorithms [14, Sec. 2]. Gluon gets its own category, due to its inherent differences to the other models, as it is not a model itself, but an ensemble model creator. Further, it utilizes decision trees, along with other base-models, for its forecasting ensemble, but also exhibits longer training-times similarly to the neural network based models.

To maintain a standardised baseline across the $1,052$ experiments performed for this paper, as detailed in section V, all models were implemented using their default configurations, with no changes to hyperparameters or loss-functions. While it is acknowledged that neural networks are highly sensitive to hyperparameter tuning and may underperform in their default states compared to tree-based algorithms, performing exhaustive hyperparameter optimization for every data constellation and horizon was computationally prohibitive. Consequently, the purpose of this paper is not to compare peak model performance against one another. The goal is to evaluate their relative performance by examining how the accuracy of different algorithmic families scales or degrades in response to varying data availability and increasing forecasting horizons. Aligning with this approach, AutoGluon was constrained to running on its fastest quality setting, trading accuracy for training speed, in order to ensure its training time would not increase prohibitively while performing walk-forward validation. To assess the impact of these simplifying choices, a supplementary set of experiments was conducted. First, the best-performing neural network model was retrained using MAE loss instead of the default MSE loss, to evaluate whether loss function alignment with the evaluation metric would improve performance. Second, hyperparameter tuning via Optuna was performed on the best, mean, and worst tree-based feature sets at the 24h horizon to quantify the improvement margin available through optimization. Third, AutoGluon was retrained at the medium_quality preset to determine whether the computational constraints of the fastest setting had materially limited its accuracy. Results from these supplementary experiments are discussed in section VI.

C. Prerequisite

As previously mentioned, the Danish electricity market is special in various ways. These circumstances will be discussed here.

1) Daily market fixing: The most important thing to note about the Danish energy market is that the hourly energy prices are fixed at noon via the day-ahead-market. At noon, Nord Pool stops accepting bids and offers, processes all agreements and trades, and then publishes the price based on this somewhere between 30-90 minutes later. After this, energy producers, consumers, and traders, may utilize the intra-day-market to minimize the fines and penalties that come with producing too much or too little depending on the actual demand. For the purposes of this paper, we only consider the day-ahead-market.

2) Regions: The Danish energy grid is divided into two distinct regions/markets: DK1 and DK2, west and east of the Great Belt respectively. These regions trade with neighbouring countries independently, as well as trade with each other. For the purposes of this paper, it simply allows us to distinguish which weather stations' measurements should be used to predict the energy prices of the appropriate region.

3) Time format: As of October 2025, Nord Pool changed their reporting format for energy prices from hourly to 15-minute increments. As this paper uses data that spans both reporting formats, we simply discard data that is not exactly on the hour, as interpolating 15-minute increments for almost 10 years of data seemed unnecessary compared to the chosen approach.

IV. Data and Preprocessing

The section will detail what data was used for training the different models, as well as how it was preprocessed, with the purpose of enabling reproducibility. All the code used for the project is available on GitHub[3]. For transparency it should be noted that generative AI tools have been used extensively for both writing and reviewing the code.

A. Data Sources

1) Weather data: The weather data used in the paper originates from DMI. They collect weather data from their numerous weather stations located in Denmark, Greenland, and the Faroe Islands. They then provide weather forecasts on a national scale.
They publish their collected data through both API and downloadable files. They also provide the data in different granularities, such as by station, municipality, country, or by 10/20 square kilometre grid areas[4]. For this paper, the data was accessed on a granularity level of individual stations. Additionally, DMI provides two different versions of meteorological

[3] link to github
[4] https://opendataapi.dmi.dk/v2/climateData/bulk/

measurement data, one version that is purely raw data, and one that has been processed for quality control by DMI employees. The quality controlled version of the data was chosen for this paper. The utilized data has a date range from 00:00:00 - 01/01/2015 until 00:00:00 - 03/25/2026, the end of the range signifies the date the download was performed. Downloading the data was preferred to avoid multiple API calls, as there were a limit of 300.000 measurements accessible through the API in each call. If one wishes to develop a prediction model utilizing this data, and update it during deployment, API calls would be the preferred method, to get limited daily data amounts.

2) Energy production & consumption data: The data regarding energy P&C originates from EN. They are a subsidiary of the Danish government and manages the Danish energy grid. The also publish a variety of data regarding the Danish energy grid, again, through both API and downloadable files. For the same reason as with DMI, downloadable files were chosen. The P&C data used for this paper is acquired from the "Production and Consumption - Settlement" publication[5].

3) Price data: The price data also originates from EN. In October of 2025 EN changed their reporting format to follow the new 15-minute timing practices of Nord Pool. Due to this, the previous publication was discontinued, and a new one was created in accordance with the new format. The price data for this paper has been concatenated between these two different publications: Elspot Prices[6], from before October 2025, and Day-Ahead-Prices[7], for after October 2025. To ensure compatibility between these two formats, we simply discard prices that are not exactly hourly in the newer dataset.

4) Midas Energy data: Midas Energy has also provided weather data in the format they use for their own prediction models. This is, in simple terms, a processed and generalised version of the DMI data. This data will later be used as a substitution for the weather data, to gauge the difference between the two varieties of weather data. It will also be added to the best performing models, to investigate its direct impact.

B. Data format and Preprocessing

This section details the data formats before and after preprocessing, as well as the preprocessing pipeline for each data source.

1) Weather data:

a) Raw data: The raw weather data as can be seen partly in table X, consists of various real-time measurements performed by a variety of instruments located at different weather stations. Each row in the raw data consists of: a measurement ID, the geographical coordinates for the measurement, the measurement parameter type, the time resolution with the related timeslots, the value, and the station ID. It also includes several features related to DMI quality assurance, such as creation time, calculation time, quality control status, etc., however, these are discarded as they have no metrological value. Further, there exists four different timestamps within each entry. For the purposes of this paper, we utilise the 'from' timestamp as the main timestamp, as that will correspond to the price from 14:00 to 15:00.

b) Preprocessed data: Preprocessing on the weather data mainly consisted of grouping each individual measurement by station ID, then aggregating and averaging measurements to become hourly if they are not already. Lastly, we once again aggregate these hourly measurements by region, using both station ID as well as measurement coordinates, to determine the region, ending up with one hourly dataset for the DK1 and DK2 regions respectively. During this geographical segregation we also excluded both Greenland, the Faroe Islands, and Bornholm, due to the geographical distance from these regions to DK1 and DK2. This resulted in data such as the example depicted in table X. Further validation of the preprocessed weather data was performed, in order to ensure data quality. Features that did not have data for the entire period were also entirely removed. This means that if a type of sensor was decommissioned during the decade of training data its feature data would be entirely excluded from training, to enable rolling-window cross-validation, as will be detailed in section V.

2) Production and Consumption data: The P&C data required minimal preprocessing, as it was already published in the hourly format. The only preprocessing performed on this dataset was dividing it into the two regions, DK1 and DK2, which was already labelled within a column of the dataset. Then we ensured that the date-range matched the DMI data. The data contained within the dataset consists of hourly production of wind energy of various size-categories of wind-farms, both off- and onshore, solar energy production of various size-categories of solar farms, fossil-fuel related energy production, exchanges with neighbouring countries, as well as Danish energy consumption data. Notably, columns relating to foreign energy exchange contained frequent missing values when no exchange had occurred between the countries during that specific hour. Since these missing values correspond to an energy exchange of zero, they were explicitly encoded as such. This ensures the models accurately interpret periods of national grid isolation, without the need for interpolation.

3) Price: The price data was divided into two publication, due to the previously mentioned Nord Pool formatting change. However, the columns in the tables where still the same, so we just ensured that they were in the same order, and then appended the most recent publication to the earlier one. We also discarded non-hourly prices to keep formatting consistent across tables.

4) General Preprocessing: Additionally, general preprocessing was performed across all datasets, or to ensure consistent UTC time formatting, also removing the issue of daylight savings time anomalies. Lastly, all datasets were sorted into ascending timestamps, and data-ranges were enforced over all datasets.

5) Master Matrix Data: After preprocessing, all of these different tables were joined by their timestamp, creating one master table, to be used for model training. Additionally,

[5]https://www.energidataservice.dk/tso-electricity/productionconsumptionsettlement

[6]https://www.energidataservice.dk/tso-electricity/Elspotprices

[7]https://www.energidataservice.dk/tso-electricity/DayAheadPrices

TABLE I: Snapshot of a Data Record (Raw vs. Preprocessed)

Raw Data Snapshot							
Timestamp	**Zone**	**Spot Price**	**Load**	**Wind Prod.**	**Solar Prod.**	**Temperature**	**Precipitation**
2026-04-26 12:00	DK1	45.20 EUR	3200 MW	12.5 MW	400 MW	14.2°C	0.0 mm
Preprocessed Data Snapshot							
Timestamp	**Zone_DK1**	**Price_Norm**	**Load_Norm**	**Wind_Norm**	**Solar_Norm**	**Temp_Norm**	**Precip_Flag**
1761476400	1	0.23	0.65	0.41	0.12	0.55	0

TABLE II: **PLACEHOLDER - NOT REAL DATA DEPICTED.**

several columns were created for the express purpose of either enhancing model performance or combatting over-fitting of the models during training. Four different categories of columns were added:

a) Date & Time: Not to be confused with date-time, or timestamps, these columns instead relate to representing the hour of the day, the day of the week, the month, and the day of the year as ascending integers. We add these to give the models an integer representation of the days and hours, to better let them interpret consumption patterns, e.g. the differing patterns present during weekdays and weekends. To enable detection of such patterns further, we also encode these integers as Sine and Cosine waves, creating a scaled version of them, that may be more useful to certain model types. Utilizing both mitigates the natural symmetrical ambiguity that would be present otherwise, mapping each time step to a unique coordinate on a unit circle. The waves were created using the following formulae 1 and 2:

$$x_{\text{sin}} = \sin\left(\frac{2\pi x}{\max(x)}\right) \tag{1}$$

$$x_{\text{cos}} = \cos\left(\frac{2\pi x}{\max(x)}\right) \tag{2}$$

where x is the integer being encoded, and $\max(x)$ is the maximum value of that specific period e.g., 24 for the hour of the day, or 365 for the day of the year. Notably, day of month was deliberately excluded as a feature, because of its variable period length, due to not all months contain the same number of days, would introduce gaps at the end of shorter months that cannot be cleanly resolved with a static denominator. A dynamic denominator, adjusting $\max(x)$ to reflect the actual number of days in each specific month, would resolve this, however, the same information is implicitly captured by the combination of the day of year and month encodings already present.

b) T-Minus data: Since we will be training tree-based models, which does not possess any inherent memory capabilities, we introduce T-Minus data, or lagging data, which are essentially just copies of earlier data at various horizons. All weather-related measurements, as well as P&C related values, have been copied and inserted 24-hours later, giving the model inherent access to yesterdays weather and consumption at this exact hour, in order to predict the price for the current hour. Additionally, for the price data specifically, t-minus data was included for the previous 24-hour, 48-hour, and 168-hour, as previous prices were deemed especially relevant for predicting the current price.

c) Targets: Originally, the absolute price was used as the target for model training. However, to combat overfitting, a new target was introduced to the master matrix: the delta target. Instead of trying to predict the absolute price, the models were instructed to predict the change in price.

$$\Delta P_{t,h} = P_{t+h} - P_t \tag{3}$$

where P_t is the price at the current hour and h is the forecast horizon in hours.
The results of both training approaches will be discussed in section VI.

d) Groupings: To make training different variations of the data easier, each column was assigned a group consisting of related columns. These groupings will be discussed further in section V.

e) Missing data and imputation: To ensure satisfactory data quality, features that are more than 90% empty, or have constant values in 95% of rows, are dropped entirely prior to training. These thresholds were chosen conservatively to retain as many features as possible, under the assumption that moderately sparse features could be recovered through imputation. A post-hoc audit confirmed that no columns were dropped under the emptiness criterion across any dataset. The constant value criterion resulted in the removal of two internal DMI quality control columns, `Stations_Reporting_Min` and `Stations_Reporting_Max`. These columns carry no meteorological value and their removal is appropriate.

For remaining missing values, three steps were performed to all columns. First, interpolation is applied to gaps of up to 12 consecutive hours, to fill in smaller gaps. Second, forward and backward filling are applied as a supplementary pass, also capped at 12 hours each, to handle edge cases where interpolation is insufficient, due to a one of the sides of the gap being open. Any values still missing after both passes, implying a consecutive gap exceeding 24 hours, are filled using the column mean as a last resort.

The behaviour of this strategy differed substantially between data sources. The price data required no imputation whatsoever, containing zero missing values across both publications. The weather data was largely clean, with 0.25% and 0.53% of values originally missing in DK1 and DK2 respectively. The vast majority of these were recovered by interpolation, with the mean fallback triggered only for `temp_soil_30` and `leaf_moisture` in specific years, accounting for at most 0.02% and 0.18% of all weather values in DK1 and DK2 respectively.

The production and consumption data presented a different situation. Approximately 9.87% of its numeric values were originally missing, concentrated entirely in four columns: `UnknownProdMWh`, `CommercialPowerMWh`, `GridLossInterconnectorsMWh`, and `GridLossDistributionMWh`. These represent production and grid measurement categories that were introduced or changed during the 11-year dataset period. In these cases, interpolation is not physically meaningful, and the column mean effectively encodes the absence of a measurement category as a neutral baseline value. Additionally, six exchange columns containing frequent missing values were zero-filled separately, as a missing exchange value physically corresponds to zero energy flow between countries rather than an unknown measurement.

We also create columns of binary flags to signal what data is natural and what is synthetic. These flags are primarily motivated by tree-based models, which lack any inherent mechanism to discount uncertain inputs. By explicitly encoding which values are synthetic, tree-based models may learn to route predictions away from relying on such criteria when other features may be more reliable. Neural network-based models can achieve a similar effect through their attention and gating mechanisms without explicit flags, however, the flags are nonetheless present within the master matrix for all models types, as they carry no cost for models that do not benefit from them.

6) Midas Energy data: No preprocessing has been performed on the Midas Energy Data, except for ensuring consistent time-formatting in relation to the other datasets.

V. Experiments

This section contains details of all the performed experiments, as well as other considerations related thereto.

A. Experiment Suite

All models were trained over data from an 11 year time-frame, from January 2015 to mid-March 2026. They were also trained in a walk-forward sliding-window with an initial training window of two years, then a test window of 30 days, as well as a step size of 30 days. As previously mentioned, data-groups were constructed in order to create different data-scenarios for training. These groups consisted of the following categories:

- **Weather**: All weather-related features.
- **WeatherLags**: T-Minus 24 hour data of the Weather group.
- **Grid**: All data related to P&C, aside from the foreign energy exchange features.
- **GridLags**: T-Minus 24 hour data of the Grid group.
- **GridExchange**: Only the data related to foreign energy exchange.
- **GridExchangeLags**: T-Minus 24 hour data of the GridExchange group.
- **Price**: Columns containing the absolute price and price delta. These are not included in the training data, but are instead individually used as prediction targets depending on the type of experiment being run, either absolute price or price delta.
- **PriceLags**: T-Minus 24, 48, and 168 hour data of both absolute price and price delta.
- **Time**: Data related to time, such as hour, day of the week, day of the month, day of the year, as well as sine and cosine representations.
- **AllFeatures**: Simply a collection of all previous groupings.

Based on these groups, we created 13 base experiments, generally increasing information sequentially:

1) Weather, Time
2) Weather, WeatherLags, Time
3) Weather, PriceLags, Time
4) Weather, WeatherLags, PriceLags, Time
5) Weather, Grid, GridExchange, Time
6) Weather, WeatherLags, Grid, GridExchange, Time
7) Weather, Grid, GridExchange, PriceLags, Time
8) Weather, WeatherLags, Grid, GridExchange, PriceLags, Time
9) Weather, Grid, GridExchange, GridLags, GridExchangeLags, Time
10) Weather, WeatherLags, Grid, GridExchange, GridLags, GridExchangeLags, Time
11) Weather, Grid, GridExchange, GridLags, GridExchangeLags, PriceLags, Time
12) Weather, WeatherLags, Grid, GridExchange, GridLags, GridExchangeLags, PriceLags, Time
13) AllFeatures

These 13 experiments were run at horizons 0, 24, 48, 72, 96, 120, 144, and 168 hour for both targets absolute price and price delta. Note that price delta was not used as a target for horizon 0 hour, as the delta between the current and target would always be 0. The 24 hour horizon was run for all eight models, with the subsequent horizons being run for all tree-based models, but only for select neural network-based models, due to their training-time constraints. This resulted in $4 \times 13 \times (7+8) = 780$ tree-based models, where 4 is the number of tree-based model types, 13 is the number of feature set configurations, and $(7 + 8)$ represents the 8 absolute price horizons and 7 price delta horizons respectively (horizon 0h is excluded from the delta target as it is always zero). For the neural network-based models and AutoGluon, an equivalent set of 4 model types were evaluated, yielding $4 \times (13 \times 2 + 3 \times 7 \times 2) = 272$ models, where the first term covers all 13 feature sets at the 24h horizon for both targets, the second term covers the 3 selected feature sets across the remaining 7 horizons and both targets, and the factor of 2 in both terms represents the two prediction targets. The reduced number of neural network-based experiments was determined by their computational constraints, with the 3 selected feature sets identified as the best, mean, and worst performers based on their 24h horizon results. This results in a total of $780 + 272 = 1052$ models trained.

B. Evaluation Criteria

While there are many different criteria one could chose to focus on when comparing the performance of models, the one chosen for this paper is Mean Absolute Error (MAE). The equation for calculating this metric is shown in Equation 4.

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n} |y_t - \hat{y}_t| \tag{4}$$

where y_t and $\hat{y}t$ are the ground truth target value at prediction time and the predicted value at prediction time respectively. In this context, this metric represents the average amount of EUR/MWh the model was off. This metric was partly chosen due to the fact that it is comparable across various model architectures and target types. Furthermore, it is expressed in the same unit as the target, making it easily interpretable. Lastly, in a market with frequent price spikes, both positive and negative, it does not penalize these extremes a harshly as other criteria. Other considered metrics are discussed in appendix A, and all of the values, both for the chosen and non-chosen metrics are available online in the GitHub repository. Additionally, a naive baseline, that simply reuses yesterday's price, calculated as follows: $\hat{y}_t = y_{t-24}$ was computed across all horizons to establish a non-ML reference point.

C. Experimental Setup

All model training, validation, and testing were conducted locally on a machine running Microsoft Windows 11 Home, equipped with a 13th Gen Intel Core i7-13650HX processor and 16 GB of RAM, utilizing CPU computation exclusively. The exclusion of the GPU was evaluated on the basis that the libraries used for training and validation of the neural network-based models were optimised for the CPU. This means that while training time was a somewhat limiting factor for the neural network-based models in this paper, this aspect may be mitigated with different training approaches and hardware, however, the performance metrics, aside from training time, should be unaffected.

D. Training, Testing & Validation

Model evaluation in this paper is performed using walk-forward validation with an expanding training window. This approach was chosen over standard k-fold cross-validation, which randomly partitions data into folds without regard for temporal ordering. Applying k-fold to time-series data would allow a model to be trained on future observations and tested on past ones, constituting a form of data leakage that would produce optimistically biased results. Walk-forward validation avoids this by strictly preserving the temporal ordering of time-series data. The model is always trained exclusively on historical data and evaluated on data it has never seen.

The procedure works as follows: An initial training window covering January 2015 through December 2016 is used to train each model for the first time. The model is then evaluated on the immediately following 30-day test window. After evaluation, the training window expands to include those 30 days, and the test window slides forward to the next 30-day period. This process repeats until the full dataset is exhausted. The expanding window design ensures that all available historical data is utilised at each step, while the fixed 30-day test window provides a consistent evaluation unit across all iterations. A figure illustrating this process is shown in Figure X. While this approach of essentially retraining the model 12 times per year, for every year in the training data, is computationally expensive, it is the generally preferred method of training when dealing with time-series data.. needs source

E. Data Leakage Prevention

The perceptive reader may have noticed that the master matrix dataset retains the raw spot price and all target columns to serve as a shared foundation for all experiments. However, feeding this matrix directly to a model would constitute severe data leakage, as the model could trivially learn to copy the target value rather than forecast it. To prevent this, a Target Shield is applied at load time before each training run, which drops all leaky columns from the feature set. Specifically, the active prediction target is removed from the features, the raw spot price is excluded except when included as a lagged historical feature, and the concurrent price delta columns are removed.

F. Feature Importance

After training various constellations and horizons of many of the models, feature importance was performed on the best performing models of each architecture of each horizon. Feature importance was determined by performing permutation importance, a method of systematically dropping each column individually and measuring the impact on your evaluation criteria. For this paper, it was deemed that if the dropping of a column resulting in even a drop of 0,01 EUR/MWh, then it would be allowed to stay. This method is an extensively rigorous, though computationally expensive, approach to feature importance. The results of these pruned models will be discussed in section VI.

G. Supplementary Experiments

To address generalisability and the limitations that come with running the models on their default settings i.e. with no hyperparameter tuning and with default loss functions, five supplementary experiments will be conducted:

a) Loss Function Alignment: : The best-performing neural network model, as will be identified from 24h baseline results, will be retrained using MAE loss instead of Mean Squared Error (MSE) to evaluate whether loss-metric alignment would improve performance.

b) Hyperparameter Tuning: : Optuna-based hyperparameter optimization will be applied to tree models on the best, mean, and worst performing feature sets at 24h for both targets, with 30 trials per configuration, to gauge the improvement such tuning may yield.

c) AutoGluon Quality Escalation: : AutoGluon will be allowed to train using the medium_quality preset and a 180 second time limit. This will be done on the best feature set at 24h, and is to be compared against the baseline fastest preset, with a six second time limit, that was used on the ordinary experiments.

d) Midas Energy Data: : Internally used weather features from Midas Energy will be merged with the master matrix and tested on all models at the 24h-horizon and on the tree-based models at all horizons, to compare against the raw DMI station-level aggregation.

e) DK2 Generalization Check: : The best tree and best neural network models will be trained on DK2 data at a 24h-horizon across best, mean, and worst feature sets to verify that DK1 findings generalize regionally.

The results for all of these supplementary experiments may be found in section reference

VI. Results

A. Summary of Findings

VII. Conclusion & Future Works

A. Conclusion

B. Future Work

To further investigate this topic, one could start with the following subjects:

a) Analysis of neighbouring countries:

b) Specialised neural networks:

c) Including alternative electricity production prices:

d) Investigation differing models on an individual weather station basis:

VIII. Acknowledgements

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APPENDIX

A. Additional evaluation criteria

Equations: In the following equations y_t and $\hat{y}t$ is the ground truth target value at prediction time and the predicted value at prediction time respectively.
Equation (5) depicts how to calculate the Root Mean Squared Error (RMSE).

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{5}$$

This metric is similar to MAE, however, it squares the errors first, to punish outliers more severely. In a market with frequent spikes, both positive and negative, this behaviour may not be ideal. While it could have been used as a performance metric, MAE was chosen instead due to their inherent similarities and its easier interpretability.

Equation (6) depicts how to calculate the Coefficient of Determination (R^2).

$$R^2 = 1 - \frac{\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}{\sum_{t=1}^{n}(y_t - \bar{y})^2} \tag{6}$$

This metric can be viewed in the same sense as an accuracy score in binary classification. As is the case with accuracy in such scenarios, it gives you a fair wholistic view of the models, but may be ineffective in capturing the intricacies of what you are actually aiming for, and therefore this metric was not chosen.

Equation (7) depicts how to calculate the Weighted Mean Absolute Percentage Error (wMAPE).

$$\text{WMAPE} = \frac{\sum_{t=1}^{n}|y_t - \hat{y}_t|}{\sum_{t=1}^{n}|y_t|} \tag{7}$$

This metric represents the errors as a percentage, which is easily understandable, however, in the case of energy markets, where prices may hover around zero, or in the negative values, for extended periods of time, this metric may become unstable, due to the low numbers involved.

Equation (8) depicts how to calculate the Symmetric Mean Absolute Percentage Error (sMAPE).

$$\text{sMAPE} = \frac{1}{n}\sum_{t=1}^{n}\frac{2|y_t - \hat{y}_t|}{|y_t| + |\hat{y}_t|} \tag{8}$$

This metric is similar to wMAPE, however it treats positive and negative errors as equally problematic, which is good in the case of price prediction, since both high-balling and low-balling the price of energy is problematic when trading energy. However, it has the some problem as wMAPE, of becoming unstable during low-number periods.

Equation (9) depicts how to calculate the Mean Directional Accuracy (MDA).

$$\text{MDA} = \frac{1}{n-1}\sum_{t=1}^{n-1}\mathbf{1}_{\text{sgn}(y_{t+1}-y_t)==\text{sgn}(\hat{y}_{t+1}-y_t)} \tag{9}$$

This metric signifies what percentage of the time the model was correct in the direction of the next price, meaning, did the price go up, or did the price go down. This is useful for energy traders, however, directionality itself is not of a measurable magnitude, and does not align with the premise of this comparison of how data constellations and horizons impact model behaviour, therefore, this metric was not chosen for this paper.